

SCAI 2026 | TRACK 1 — SMART SENSING & INTELLIGENT ELECTRONIC SYSTEMS

DisasterShield

AI-Powered Multi-Hazard Early Warning System with Edge Intelligence

Connected Intelligence for Climate Resilience

Harsh C. Parmar

IEEE Student Branch, SVNIT Surat

The Problem

Climate change is outpacing our ability to warn communities in time



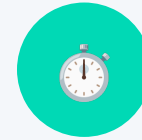
Recurring hazards

Floods, wildfires, droughts across India



High economic loss

Crops, property & livelihoods at risk yearly



Too slow to act

Centralized systems can't give local alerts

*Existing early-warning systems are expensive, centralized, and cannot deliver the localized, real-time alerts that **vulnerable rural communities and farmers actually need.***

Proposed Solution

A low-cost, edge-AI system that watches, senses, and warns — in real time



Real-Time Vision

Trained AI model detects fire & smoke from live camera feed



Sensor Fusion

Combines vision with temperature, humidity, rain & soil data



Edge-Ready

TensorFlow Lite model runs offline, no cloud dependency

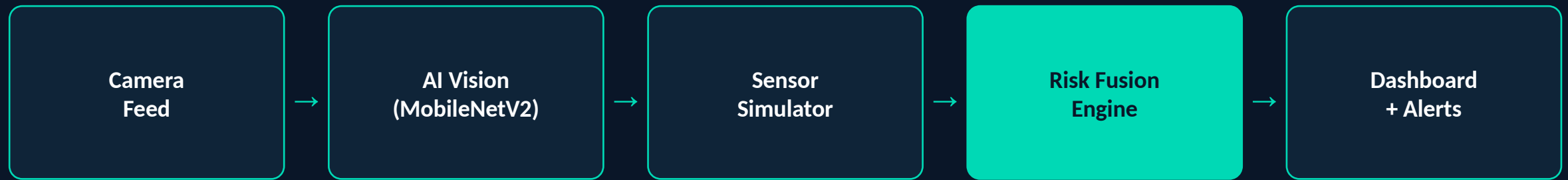


Multi-Hazard

Single risk engine covers fire, flood & drought signals

System Architecture

From live camera frame to actionable alert



Vision confidence and simulated sensor trends are combined by the Fusion Engine into one Low / Medium / High risk level, shown live on the dashboard.

AI & Computer Vision

MobileNetV2 transfer learning, trained on a real fire/non-fire dataset

99%

Training Accuracy

97.99%

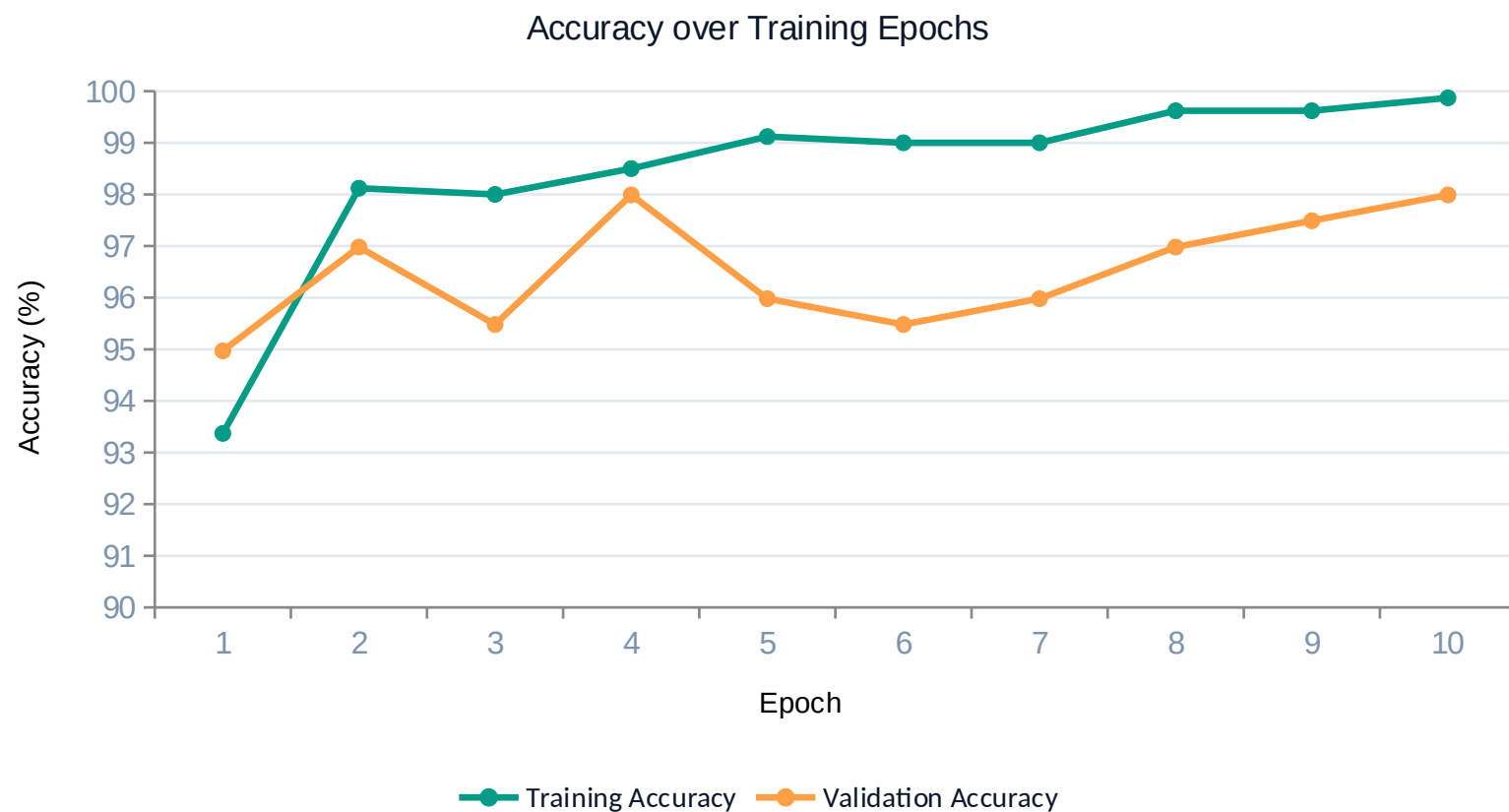
Validation Accuracy

10

Training Epochs

9.2 MB

TFLite Model Size



Live Demo

Watch the system detect fire/smoke and raise an alert in real time

AI-Powered Real-time Fire Detection (MobileNetV2)



1

Live camera feed

Streamed via Flask, processed frame-by-frame

2

AI confidence score

MobileNetV2 model output, updated every frame

3

Instant risk label

NORMAL / MEDIUM / HIGH RISK shown in real time

Multi-Hazard Risk Engine

Vision confidence + simulated sensors fused into one risk level

Temperature

Humidity

Smoke Level

Rainfall

Soil Moisture

↓ fused with AI vision confidence ↓

HIGH RISK

Fire confidence high or smoke/temperature spike — immediate action required

MEDIUM RISK

Elevated readings, borderline confidence — monitor closely

LOW RISK

Normal readings across vision and sensors — conditions stable

Technologies & Implementation

Built entirely in Python, edge-deployment ready



Python 3.13

Core application language



OpenCV

Camera capture & frame processing



TensorFlow Lite

Edge-optimized AI inference



Flask

Live web dashboard



Open Source on GitHub

github.com/HarshParmar029/DisasterShield-SCAI2026

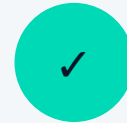
Edge-ready today — designed to run on Raspberry Pi 4/5 with the same TFLite model.

Current Achievements

What's built, tested, and working today — not just planned



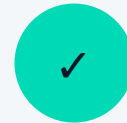
Working live dashboard with real-time camera feed



Trained AI model — 97.99% validation accuracy



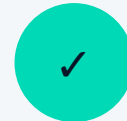
Real-time multi-hazard risk fusion engine



Complete GitHub repository with training notebook



Confusion matrix & sample predictions validated



TensorFlow Lite export — ready for edge deployment

Results & Impact

A working system with numbers to back it up

97.99%

Validation accuracy on held-out fire/non-fire images

< 1 sec

Detection latency per frame on a laptop CPU

9.2 MB

Compact TFLite model — fits on edge hardware

Societal Impact

Faster, localized alerts mean rural communities and farmers can protect lives, crops, and property before a hazard escalates — without expensive centralized infrastructure.

Future Scope

From prototype to field-deployable system

1

Real Hardware Sensors

Deploy on Raspberry Pi with DHT22, MQ-135 and soil sensors

2

Remote Sensing

Add NDVI/satellite-based wide-area risk mapping (Track 3 tie-in)

3

Connectivity

LoRaWAN for long-range, low-power alert transmission

4

Mobile App

Push notifications for field teams and affected communities

Connected Intelligence

for a Safer, More Resilient India

Thank you.

Questions & Discussion

Harsh C. Parmar • IEEE SVNIT Surat

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